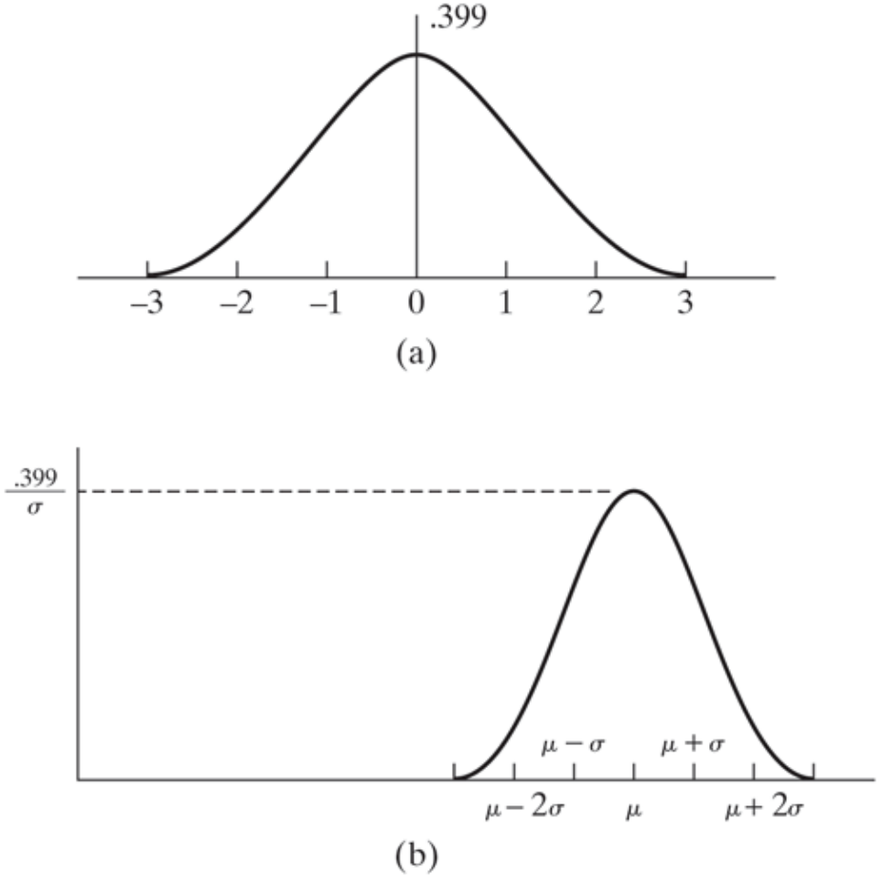
5.4 Normal Random Variables

We say that X is a normal random variable, or simply that X is normally distributed, with parameters μ and σ2 if the density of X is given by

$$f(x)=\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2} \qquad -\infty<x<\infty$$

This density function is a bell-shaped curve that is symmetric about μ. (See **Figure 5.5**.)

Figure 5.5 Normal density function: (a) μ=0, σ=1; (b) arbitrary μ, σ2.



The normal distribution was introduced by the French mathematician Abraham DeMoivre in 1733, who used it to approximate probabilities associated with binomial random variables when the binomial parameter n is large. This result was later extended by Laplace and others and is now encompassed in a probability theorem known as the central limit theorem, which is discussed in **Chapter 8**. The *central limit theorem*, one of the two most important results in probability theory,† gives a theoretical base to the often noted empirical observation that, in practice, many random phenomena obey, at least approximately, a normal probability distribution. Some examples of random phenomena obeying this behavior are the height of a man or woman, the velocity in any direction of a molecule in gas, and the error made in measuring a physical quantity.

† The other is the strong law of large numbers.

To prove that f(x) is indeed a probability density function, we need to show that

$$\frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty}e^{-(x-\mu)^2/2\sigma^2}\,dx=1$$

Making the substitution $y=(x-\mu)/\sigma$, we see that

$$\frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} e^{-(x-\mu)^2/2\sigma^2}\,dx=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-y^2/2}\,dy$$

Hence, we must show that

$$\int_{-\infty}^{\infty} e^{-y^2/2}\,dy=\sqrt{2\pi}$$

Toward this end, let $I=\int_{-\infty}^{\infty} e^{-y^2/2}\,dy$. Then

$$I^2=\int_{-\infty}^{\infty} e^{-y^2/2}\,dy\int_{-\infty}^{\infty} e^{-x^2/2}\,dx=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(y^2+x^2)/2}\,dy\,dx$$

We now evaluate the double integral by means of a change of variables to polar coordinates. (That is, let $x=r\cos\theta$, $y=r\sin\theta$, and $dy\,dx=r\,d\theta\,dr$.) Thus,

$$I^2=\int_0^{\infty}\int_0^{2\pi} e^{-r^2/2}r\,d\theta\,dr=2\pi\int_0^{\infty} re^{-r^2/2}\,dr=-2\pi e^{-r^2/2}\Big|_0^{\infty}=2\pi$$

Hence, $I=\sqrt{2\pi}$, and the result is proved.

An important fact about normal random variables is that if X is normally distributed with parameters μ and σ^2, then $Y=aX+b$ is normally distributed with parameters $a\mu+b$ and $a^2\sigma^2$. To prove this statement, suppose that $a>0$. (The proof when $a<0$ is similar.) Let F_Y denote the cumulative distribution function of Y. Then

$$F_Y(x)=P\{Y\le x\}=P\{aX+b\le x\}=P\{X\le x-ba\}=F_X(x-ba)$$

where F_X is the cumulative distribution function of X. By differentiation, the density function of Y is then

$$f_Y(x)=\frac{1}{a}f_X(x-ba)=\frac{1}{\sqrt{2\pi}a\sigma}\exp\{-(x-ba-\mu)^2/2\sigma^2\}=\frac{1}{\sqrt{2\pi}a\sigma}\exp\{-(x-b-a\mu)^2/2(a\sigma)^2\}$$

which shows that Y is normal with parameters $a\mu+b$ and $a^2\sigma^2$.

An important implication of the preceding result is that if X is normally distributed with parameters μ and σ^2, then $Z=(X-\mu)/\sigma$ is normally distributed with parameters 0 and 1. Such a random variable is said to be a *standard*, or a *unit*, normal random variable.

We now show that the parameters μ and σ^2 of a normal random variable represent, respectively, its expected value and variance.

Example 4a

Find E[X] and Var(X) when X is a normal random variable with parameters μ and σ^2.

Solution

Let us start by finding the mean and variance of the standard normal random variable $Z=(X-\mu)/\sigma$. We have

$$E[Z]=\int_{-\infty}^{\infty} xf_Z(x)\,dx=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} xe^{-x^2/2}\,dx=-\frac{1}{\sqrt{2\pi}}e^{-x^2/2}\Big|_{-\infty}^{\infty}=0$$

Thus,

$$\mathrm{Var}(Z)=E[Z^2]=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} x^2e^{-x^2/2}\,dx$$

Integration by parts (with $u=x$ and $dv=xe^{-x^2/2}$) now gives

$$\mathrm{Var}(Z)=\frac{1}{\sqrt{2\pi}}\left(-xe^{-x^2/2}\Big|_{-\infty}^{\infty}+\int_{-\infty}^{\infty} e^{-x^2/2}\,dx\right)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-x^2/2}\,dx=1$$

Because $X=\mu+\sigma Z$, the preceding yields the results

$$E[X]=\mu+\sigma E[Z]=\mu$$

and

$$Var(X)=\sigma^2 Var(Z)=\sigma^2$$

It is customary to denote the cumulative distribution function of a standard normal random variable by $\Phi(x)$. That is,

$$\Phi(x)=\frac{1}{2\pi}\int_{-\infty}^{x} e^{-y^2/2}\,dy$$

The values of $\Phi(x)$ for nonnegative x are given in **Table 5.1**. For negative values of x, $\Phi(x)$ can be obtained from the relationship

$$\Phi(-x)=1-\Phi(x) \qquad -\infty<x<\infty \tag{4.1}$$

Table 5.1 Area Φ(x) Under the Standard Normal Curve to the Left of X.

X	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

The proof of **Equation (4.1)**, which follows from the symmetry of the standard normal density, is left as an exercise. This equation states that if Z is a standard normal random variable, then

$$P\{Z \le -x\} = P\{Z > x\} \qquad -\infty < x < \infty$$

Since $Z = (X - \mu)/\sigma$ is a standard normal random variable whenever X is normally distributed with parameters μ and σ^2, it follows that the distribution function of X can be expressed as

$$F_X(a) = P\{X \le a\} = P\left(\frac{X - \mu}{\sigma} \le \frac{a - \mu}{\sigma}\right) = \Phi\left(\frac{a - \mu}{\sigma}\right)$$

Example 4b

If X is a normal random variable with parameters $\mu = 3$ and $\sigma^2 = 9$, find (a) $P\{2 < X < 5\}$; (b) $P\{X > 0\}$; (c)

$P\{|X-3|>6\}$.

Solution

a. $P\{2<X<5\}=P\{2-33<X-33<5-33\}=P\{-13<Z<23\}=\Phi(23)-\Phi(-13)=\Phi(23)-[1-\Phi(13)]\approx.3779$

b. $P\{X>0\}=P\{X-33>0-33\}=P\{Z>-1\}=1-\Phi(-1)=\Phi(1)\approx.8413$

c. $P\{|X-3|>6\}=P\{X>9\}+P\{X<-3\}=P\{X-33>9-33\}+P\{X-33<-3-33\}=P\{Z>2\}+P\{Z<-2\}=1-\Phi(2)+\Phi(-2)=2[1-\Phi(2)]\approx.0456$

Example 4c

An examination is frequently regarded as being good (in the sense of determining a valid grade spread for those taking it) if the test scores of those taking the examination can be approximated by a normal density function. (In other words, a graph of the frequency of grade scores should have approximately the bell-shaped form of the normal density.) The instructor often uses the test scores to estimate the normal parameters μ and $\sigma 2$ and then assigns the letter grade A to those whose test score is greater than $\mu+\sigma$, B to those whose score is between μ and $\mu+\sigma$, C to those whose score is between $\mu-\sigma$ and μ, D to those whose score is between $\mu-2\sigma$ and $\mu-\sigma$, and F to those getting a score below $\mu-2\sigma$. (This strategy is sometimes referred to as grading “on the curve.”) Since

$P\{X>\mu+\sigma\} =P\{ \frac{X-\mu}{\sigma} >1 \}=1-\Phi(1)\approx .1587$ $P\{\mu<X<\mu+\sigma\} =P\{ 0< \frac{X-\mu}{\sigma} <1 \}=\Phi(1)-\Phi(0)\approx .3413$ $P\{\mu-\sigma<X<\mu\} =P\{ -1< \frac{X-\mu}{\sigma} <0 \} =\Phi(0)-\Phi(-1)\approx .3413$ $P\{\mu-2\sigma<X<\mu-\sigma\} =P\{ -2< \frac{X-\mu}{\sigma} <-1 \} =\Phi(2)-\Phi(1)\approx .1359$ $P\{X<\mu-2\sigma\} =P\{ \frac{X-\mu}{\sigma} <-2 \}=\Phi(-2)\approx .0228$

it follows that approximately 16 percent of the class will receive an A grade on the examination, 34 percent a B grade, 34 percent a C grade, and 14 percent a D grade; 2 percent will fail.

Example 4d

An expert witness in a paternity suit testifies that the length (in days) of human gestation is approximately normally distributed with parameters $\mu=270$ and $\sigma 2=100$. The defendant in the suit is able to prove that he was out of the country during a period that began 290 days before the birth of the child and ended 240 days before the birth. If the defendant was, in fact, the father of the child, what is the probability that the mother could have had the very long or very short gestation indicated by the testimony?

Solution

Let X denote the length of the gestation, and assume that the defendant is the father. Then the probability that the birth could occur within the indicated period is

$P\{X>290 \text{ or } X<240\}=P\{X>290\}+P\{X<240\}=P\{\frac{X-270}{10}>2\}+P\{\frac{X-270}{10}<-3\}=1-\Phi(2)+1-\Phi(3)\approx.0241$

Example 4e

Suppose that a binary message either 0 or 1 must be transmitted by wire from location A to location B. However, the data sent over the wire are subject to a channel noise disturbance, so, to reduce the possibility of error, the value 2 is sent over the wire when the message is 1 and the value -2 is sent when the message is 0. If $x, x=\pm 2$, is the value sent at location A, then R, the value received at location B, is given by $R=x+N$, where N is the channel noise disturbance. When the message is received at location B, the receiver decodes it according to the following rule:

If $R\geq.5$, then 1 is concluded.

If $R<.5$, then 0 is concluded.

Because the channel noise is often normally distributed, we will determine the error probabilities when N is a

standard normal random variable.

Two types of errors can occur: One is that the message 1 can be incorrectly determined to be 0, and the other is that 0 can be incorrectly determined to be 1. The first type of error will occur if the message is 1 and $2+N<.5$, whereas the second will occur if the message is 0 and $-2+N\geq.5$. Hence,

$$P\{\text{error}|\text{message is 1}\}=P\{N<-1.5\}=1-\Phi(1.5)\approx.0668$$

and

$$P\{\text{error}|\text{message is 0}\}=P\{N\geq 2.5\}=1-\Phi(2.5)\approx.0062$$

Example 4f

Value at Risk (VAR) has become a key concept in financial calculations. The VAR of an investment is defined as that value v such that there is only a 1 percent chance that the loss from the investment will be greater than v. If X, the gain from an investment, is a normal random variable with mean μ and variance σ^2, then because the loss is equal to the negative of the gain, the VAR of such an investment is that value v such that

$$.01=P\{-X>v\}$$

Using that $-X$ is normal with mean $-\mu$ and variance σ^2, we see that

$$.01=P\{-X+\mu\sigma>v+\mu\sigma\}=1-\Phi(v+\mu\sigma)$$

Because, as indicated by **Table 5.1**, $\Phi(2.33)=.99$, we see that

$$v+\mu\sigma=2.33$$

That is,

$$v=\text{VAR}=2.33\sigma-\mu$$

Consequently, among a set of investments all of whose gains are normally distributed, the investment having the smallest VAR is the one having the largest value of $\mu-2.33\sigma$.

5.4.1 The Normal Approximation to the Binomial Distribution

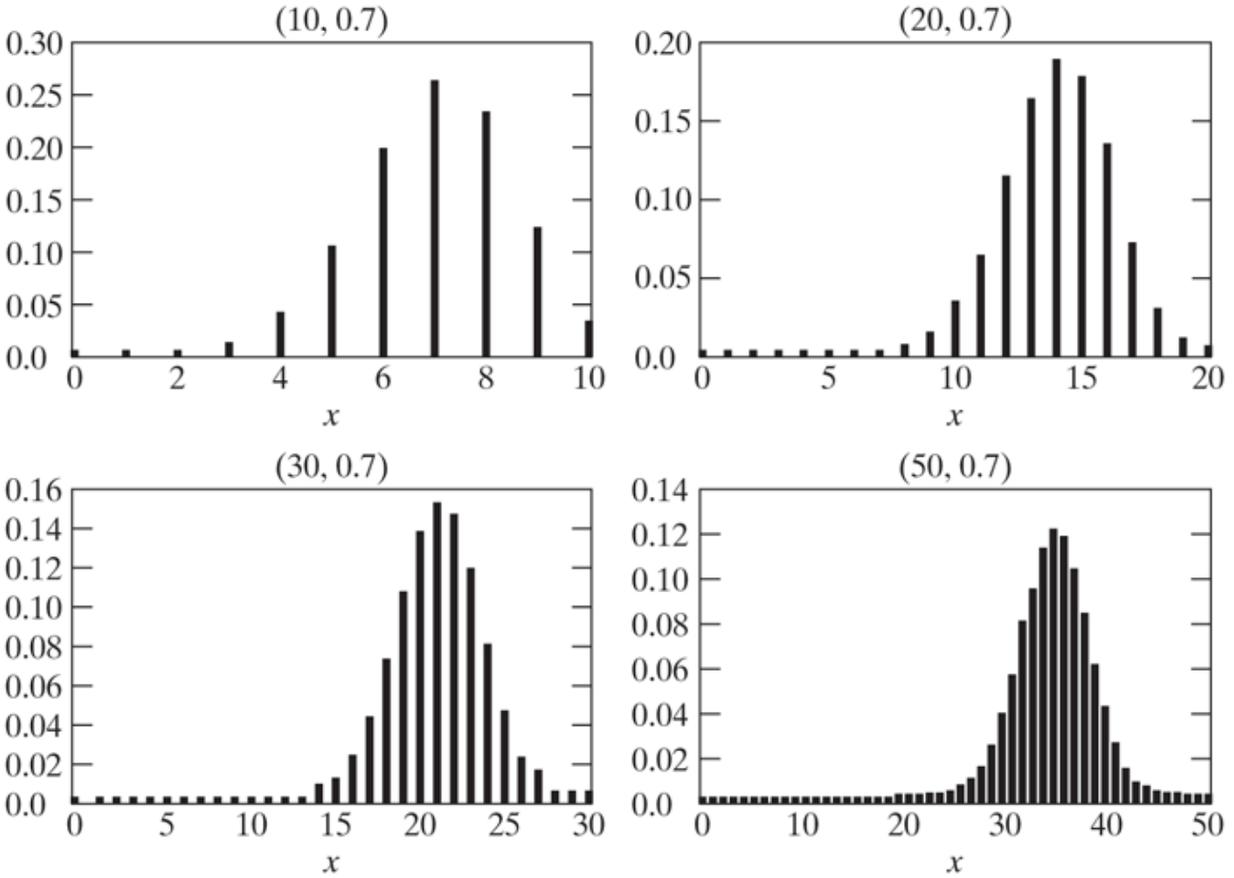
An important result in probability theory known as the DeMoivre–Laplace limit theorem states that when n is large, a binomial random variable with parameters n and p will have approximately the same distribution as a normal random variable with the same mean and variance as the binomial. This result was proved originally for the special case of $p=\frac{1}{2}$ by DeMoivre in 1733 and was then extended to general p by Laplace in 1812. It formally states that if we "standardize" the binomial by first subtracting its mean *np* and then dividing the result by its standard deviation $\sqrt{np(1-p)}$, then the distribution function of this standardized random variable (which has mean 0 and variance 1) will converge to the standard normal distribution function as $n\to\infty$.

> The DeMoivre–Laplace Limit Theorem
>
> If S_n denotes the number of successes that occur when n independent trials, each resulting in a success with probability p, are performed, then, for any $a<b$,
>
> $$P\left\{a\leq\frac{S_n-np}{\sqrt{np(1-p)}}\leq b\right\}\to\Phi(b)-\Phi(a)$$
>
> as $n\to\infty$.

Because the preceding theorem is only a special case of the central limit theorem, which is presented in **Chapter 8**, we shall not present a proof.

Note that we now have two possible approximations to binomial probabilities: the Poisson approximation, which is good when n is large and p is small, and the normal approximation, which can be shown to be quite good when np(1−p) is large. (See **Figure 5.6**.) [The normal approximation will, in general, be quite good for values of n satisfying np(1−p)≥10.]

Figure 5.6 The probability mass function of a binomial (*n, p*) random variable becomes more and more "normal" as *n* becomes larger and larger.



Example 4g

Let X be the number of times that a fair coin that is flipped 40 times lands on heads. Find the probability that X=20. Use the normal approximation and then compare it with the exact solution.

Solution

To employ the normal approximation, note that because the binomial is a discrete integer-valued random variable, whereas the normal is a continuous random variable, it is best to write P{X=i} as P{i−1/2<X<i+1/2} before applying the normal approximation (this is called the *continuity correction*). Doing so gives

$$P\{X=20\}=P\{19.5<X<20.5\}=P\{19.5-2010<X-2010<20.5-2010\}\approx P\{-.16<X-2010<.16\}\approx\Phi(.16)-\Phi(-.16)\approx.1272$$

The exact result is

$$P\{X=20\}=(4020)(12)40\approx.1254$$

Example 4h

The ideal size of a first-year class at a particular college is 150 students. The college, knowing from past experience that, on the average, only 30 percent of those accepted for admission will actually attend, uses a policy of approving the applications of 450 students. Compute the probability that more than 150 first-year

students attend this college.

Solution

If X denotes the number of students who attend, then X is a binomial random variable with parameters n=450 and p=.3. Using the continuity correction, we see that the normal approximation yields

$$P\{X \geq 150.5\} = P\{X - (450)(.3) \over 450(.3)(.7) \geq 150.5 - (450)(.3) \over 450(.3)(.7)\} \approx 1 - \Phi(1.59) \approx .0559$$

Hence, less than 6 percent of the time do more than 150 of the first 450 accepted actually attend. (What independence assumptions have we made?)

Example 4i

To determine the effectiveness of a certain diet in reducing the amount of cholesterol in the bloodstream, 100 people are put on the diet. After they have been on the diet for a sufficient length of time, their cholesterol count will be taken. The nutritionist running this experiment has decided to endorse the diet if at least 65 percent of the people have a lower cholesterol count after going on the diet. What is the probability that the nutritionist endorses the new diet if, in fact, it has no effect on the cholesterol level?

Solution

Let us assume that if the diet has no effect on the cholesterol count, then, strictly by chance, each person's count will be lower than it was before the diet with probability 12. Hence, if X is the number of people whose count is lowered, then the probability that the nutritionist will endorse the diet when it actually has no effect on the cholesterol count is

$$\sum_{i=65}^{100} \binom{100}{i} \left(\frac{1}{2}\right)^{100} = P\{|X| \geq 64.5\} = P\left\{ \frac{X - (100)\left(\frac{1}{2}\right)}{100\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)} \geq 2.9 \right\} \approx 1 - \Phi(2.9) \approx .0019$$

Example 4j

Fifty-two percent of the residents of New York City are in favor of outlawing cigarette smoking on university campuses. Approximate the probability that more than 50 percent of a random sample of n people from New York are in favor of this prohibition when

a. n=11
b. n=101
c. n=1001

How large would n have to be to make this probability exceed .95?

Solution

Let N denote the number of residents of New York City. To answer the preceding question, we must first understand that a *random sample* of size n is a sample such that the n people were chosen in such a manner that each of the $\binom{N}{n}$ subsets of n people had the same chance of being the chosen subset. Consequently, Sn, the number of people in the sample who are in favor of the smoking prohibition, is a hypergeometric random variable. That is, Sn has the same distribution as the number of white balls obtained when n balls are chosen from an urn of N balls, of which .52N are white. But because N and .52N are both large in comparison with the sample size n, it follows from the binomial approximation to the hypergeometric (see **Section 4.8.3**) that the distribution of Sn is closely approximated by a binomial distribution with parameters n and p=.52. The normal approximation to the binomial distribution then shows that

$$P\{Sn > .5n\} = P\left\{ \frac{Sn - .52n}{n(.52)(.48)} > \frac{.5n - .52n}{n(.52)(.48)} \right\} = P\left\{ \frac{Sn - .52n}{n(.52)(.48)} > \frac{.04n}{} \right\} \approx \Phi(.04n)$$

Thus,

$$P\{Sn > .5n\} \approx \begin{cases} \Phi(.1328) = .5528, & \text{if } n = 11 \\ \Phi(.4020) = .6562, & \text{if } n = 101 \\ \Phi(1.2665) = .8973, & \text{if } n = 1001 \end{cases}$$

In order for this probability to be at least .95, we would need $\Phi(.04n)>.95$. Because $\Phi(x)$ is an increasing function and $\Phi(1.645)=.95$, this means that

$.04n>1.645$

or

$n\geq 1691.266$

That is, the sample size would have to be at least 1692.

Historical notes concerning the normal distribution

The normal distribution was introduced by the French mathematician Abraham DeMoivre in 1733. DeMoivre, who used this distribution to approximate probabilities connected with coin tossing, called it the exponential bell-shaped curve. Its usefulness, however, became truly apparent only in 1809, when the famous German mathematician Karl Friedrich Gauss used it as an integral part of his approach to predicting the location of astronomical entities. As a result, it became common after this time to call it the *Gaussian distribution*.

During the mid- to late 19th century, however, most statisticians started to believe that the majority of data sets would have histograms conforming to the Gaussian bell-shaped form. Indeed, it came to be accepted that it was "normal" for any well-behaved data set to follow this curve. As a result, following the lead of the British statistician Karl Pearson, people began referring to the Gaussian curve by calling it simply the *normal* curve. (A partial explanation as to why so many data sets conform to the normal curve is provided by the central limit theorem, which is presented in **Chapter 8**.)

Abraham DeMoivre (1667–1754)

Today there is no shortage of statistical consultants, many of whom ply their trade in the most elegant of settings. However, the first of their breed worked, in the early years of the 18th century, out of a dark, grubby betting shop in Long Acres, London, known as Slaughter's Coffee House. He was Abraham DeMoivre, a Protestant refugee from Catholic France, and, for a price, he would compute the probability of gambling bets in all types of games of chance.

Although DeMoivre, the discoverer of the normal curve, made his living at the coffee shop, he was a mathematician of recognized abilities. Indeed, he was a member of the Royal Society and was reported to be an intimate of Isaac Newton.

Listen to Karl Pearson imagining DeMoivre at work at Slaughter's Coffee House: "*I picture DeMoivre working at a dirty table in the coffee house with a broken-down gambler beside him and Isaac Newton walking through the crowd to his corner to fetch out his friend. It would make a great picture for an inspired artist*."

Karl Friedrich Gauss

Karl Friedrich Gauss (1777–1855), one of the earliest users of the normal curve, was one of the greatest mathematicians of all time. Listen to the words of the well-known mathematical historian E. T. Bell, as expressed in his 1954 book *Men of Mathematics*: In a chapter entitled "The Prince of Mathematicians," he writes, "*Archimedes, Newton, and Gauss; these three are in a class by themselves among the great mathematicians, and it is not for ordinary mortals to attempt to rank them in order of merit. All three started tidal waves in both pure and applied mathematics. Archimedes esteemed his pure mathematics more highly than its applications;*

Newton appears to have found the chief justification for his mathematical inventions in the scientific uses to which he put them; while Gauss declared it was all one to him whether he worked on the pure or on the applied side."